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Examination paper for MOL3100 Introduction to Molecular Medicine

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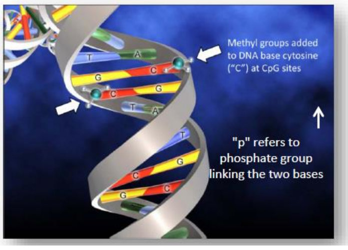
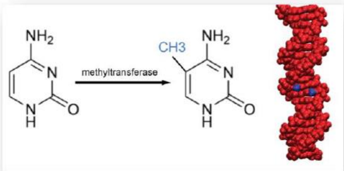
Question 1 (13p)
Cells use several mechanisms to regulate gene expression, for example, by the action of transcription factors that activates or represses transcription by binding to gene promoters and enhancers. In addition, epigenetic modifications and non-coding RNAs are important in regulation of gene activity and drive cell differentiation.

A. Epigenetic modifications occur both at the DNA level and at the protein level. Describe DNA methylation and indicate how it may regulate transcription. (2p)

DNA methylation

Cytosine can be methylated
- Stable modification, best studied epigenetic mark
- DNA methylation occurs at **C** in a **CpG** context (C before G in the DNA sequence)
- 70-80% of all CpG sites in the genome is methylated (CpGs are underrepresented in the genome)
- Silencing of genes (different methylation pattern in embryonic and somatic cells)
- In females: Inactive X chromosome is heavily DNA methylated





DNA methylation; regulation of gene expression

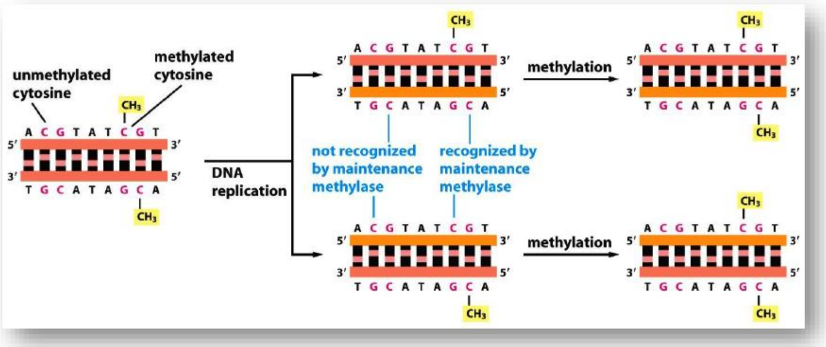
What causes the silencing effect of DNA methylation?
- Methylation of DNA **does not affect the charge.**
- The methyl group may physically prevent the binding of transcription factors.
- Methylated DNA recruits proteins that have a **methyl-CpG-binding domain**, and these proteins will in turn **recruit** other chromatin remodeling factors.

→ **Methylated DNA function as a signal for other proteins to locate to the locus and silence genes and form heterochromatin.**

B. DNA methylation patterns are usually transferred from the mother cell to the daughter cell during cell division. How does this occur? (1p)

DNA methyltransferase 1 (DNMT1) acts preferentially on those CG sequences that are base-paired with a CG sequence that is already methylated. As a result, the pattern of DNA methylation on the parental DNA strand serves as a template for the methylation of the daughter DNA strand, causing this pattern to be inherited directly following DNA replication.

Maintenance DNA methylation by DNMT1



CpG methylation is inherited from mother cell to daughter cells

→ Maintenance DNA methyltransferases

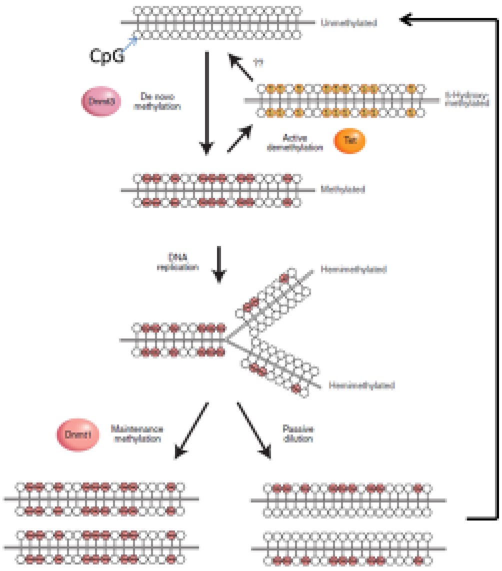
→ Stable modification

C. Although epigenetic marks are inherited, epigenetic modifications may also be dynamically regulated at specific loci. Describe the difference between passive and active DNA demethylation. (2p)

DNA demethylation; active and passive mechanisms

Active demetylation: enzymatic prosess TET mediated oxidation of the methyl-group followed by base excision repair (BER)

Passive demethylation : lack of maintenance methylation during replication, no DNMT1 or inhibition of DNMT1



DNA methylation in mammals.
Li E, Zhang Y., Cold Spring Harb Perspect Biol. 2014 May 1

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D. Genome sequencing of many cancers has revealed that genes involved in epigenetic regulation are frequently mutated. Discuss shortly how alterations in epigenetic modifications can drive cancer development. (2p) Dysregualtion of gene expression. Upregulation of oncogenes. Scilencing of tumoursupressors (most important !)

E. Non-coding RNAs (ncRNAs) are usually devided into long ncRNAs (>200 nt) and small ncRNA (<200 nt) including micro RNA (miRNA). How do miRNAs regulate gene expression? (2p) Guides RISC complex to mRNA through basepairing with 3'UTR. Depending on the degree of complimentarity in the binding, this will cause either degradation of the mRNA, mRNA cleavage, or inhibition of protein synthesis from the targeted mRNA. miRNAs thereby regulate gene expression post-transcriptionally. One miRNA can target many mRNAs, and one mRNA can be targeted by many miRNAs.

F. Discuss shortly how non-coding RNAs can contribute to disease. Provide examples. (2p)

Aberrant expression levels of ncRNAs can shift the chemical equilibrium in cells, thereby causing disease.Failure of binding to intended protein partner in ribonucleic complexes. This could be due to mutations in the ncRNA, or in the gene encoding the protein binding partner.

Examples: lncRNA in cancer (e.g. HOTAIR) and Alzheimer's disease (e.g. BACE1-AS) and has altered expression in a range of diseases, miRNAs act as oncogenes (e.g. miR-17/91 cluster) and has altered expression in a range of diseases, piRNAs in infertility (PIWI mutation) and cancers (altered expression), snoRNA in cancers and Prader-Willi syndrome, snRNAs in dwarfism and spinal muscular atrophy, etc.

G. RNA molecules can be used as therapeutic agents. Mention some benefits and challenges of using RNA molecules as medicine. (2 p)

<u>Benefits</u>

- Can affect protein expression and fix some errors in protein production
- Examples of potential targets: Pain receptors, inhibition of harmful genes, inhibition of viruses
- Easy and cheap to synthesize
- Can be used against "undruggable targets" (even noncoding genes)
- So far no evidence of adaptive immune response
- Easy to combine
- Suitable for personalized medicine

<u>Challenges</u>

- Intracellular delivery across cell and endosomal membranes
- Poor pharmacokinetic properties, partly due to urinary excretion and ubiquitous RNases
- Activation of innate immune nucleic acid sensors
- Off-target effects:
  - ✓ Suppression of unintended homologous targets
  - ✓ Activation of DNA repair pathways
  - ✓ Translocations or imprecise gene editin

- **Question 2 (8p)**

A 39-year-old Norwegian man living in Trondheim visits his medical doctor for a routine health check. After a weak he gets a call from his doctor, and is told that he has too much cholesterol in the blood (total cholesterol 10.7 mmol/l; reference values : 3.3 – 6.9 mmol/l). The doctor recommends the man to have a genetic test to either confirm or exclude that he may suffer from familial hypercholesterolemia, which is a heritable condition.

A. What is the most common cause for familial hypercholesterolemia (which protein is most commonly affected), and what inheritance pattern does this condition display? (1p) Answer: Familial hypercholesterolemia is most commonly caused by mutations that affect the normal functioning of the LDL receptor. Autosomal dominant inheritance.

B. The man agrees to have the test. After a while, he receives the result, which shows a non-sense mutation that most probably explains his high cholesterol level. What is meant by non-sense mutation? (1p) Answer: A non-sense mutation involves the change of an amino acid codon into a stop codon.

C. The man gets perplexed about the result and asks if other members in his family may also have the condition. He has a son and a daughter, who are twins, and is anxious that they may have the mutation. The doctor tells him that this is a possibility, but he cannot tell exactly what the risk is because he has no information on the genotype of the children's mother. What is the chance that neither of the twins has inherited the mutation from their father when he is heterozygote? (1p) Answer: The chance of not inheriting the mutation is 0.5. Thus the chance that neither of the twins (who have to be dizygotic since there is a boy and a girl, and who genetically can be considered as siblings) will inherit the mutation is: 0.5 x 0.5 = **0.25** (or in percent 50% x 50% = 25%).

D. Familial hypercholesterolemia is a quite common condition in Norway. What is the approximate estimated risk that his wife, who is also a Norwegian, also has the condition? (1p) Answer: Familial hypercholesterolemia is the most common autosomal dominant disorder in Norway. Her risk will be equal to the population prevalence, which is estimated to be 1:350 – 1:500.

E. He asks the doctor if it is certain that if you have the mutation - you will develop high cholesterol levels. This condition, the doctor says, displays 100% penetrance, but may show some variable expressivity due to possible modifier genes. What is meant by penetrance and variable expressivity in this context? (1p) Answer: **Penetrance** is the fraction (%) of individuals carrying a specific mutation who will develop the disease. E.g. if 80 out of 100 with the mutation develop the disease, the penetrance of that disease is 80%. **Variable expressivity** refers to variation in the severity of the disease among individuals carrying the same mutation.

F. The doctor invites the man for a talk about the possibilities of effective treatment to lower his cholesterol levels. This is important, he says, to avoid the development of atherosclerosis. Describe briefly the main features of the different stages of atherosclerosis development. (3p) Answer:

1. **The fatty streak formation (initiation):** - Endothelial cell dysfunction - Lipoprotein entry and modification - Leukocyte recruitment and foam cell formation

2. **Plaque progression (adaption)** : Smooth muscle cell (SMC) migration - Extracellular matrix metabolism

3. **Plaque disruption (clinical manifestiation) :** Integrity of plaques - Thrombogenic potential

Question 3 (6p)

The majority of virus and bacteria is harmless to humans, but some are patogenic and cause disease. One such example is Influenza A virus.

A. Influenza A viruses are classified into subtypes. What viral elements are used for this classification? (1p)

Answer:

Influenza A viruses are classified according to their surface proteins hemagglutinin (H1-16) and neuraminidase (N1-9).

B. Influenza viruses change via two different mechanisms, termed antigenic drift and antigenic shift. Explain each of these two mechanisms, and give examples. (2p)

Answer:

Antigenic drift is caused by the high error rate of the viral polymerase. Mutations emerge in each replication cycle, some of which will modify the NA or HA surface proteins, so that they escape the immune system and infect persons that are immune to the pre-existing strain. In some cases, it can change the viral tropism to other mammals. This happens every year and is the cause to why the vaccine has to be updated every year.

Antigenic shift happens when there is reassortment of segments of the viral genome. This can produce a virus with completely new properties, and is the cause of pandemics.

C. Bactria are the most abundant form of cellular life on earth. There are likely more bacterial cells than human cells in the human body where they have an important role in the gut flora. Why are some bacteria pathogenic to humans? (2p)

Answer:

1) Ability to adhere to and colonize skin or mucosal surfaces, 2) Ability to cross skin or mucosal barriers, 3) Ability to cause tissue destruction, 4) Ability to produce/secrete toxins, 5) Ability to cause inflammation, and/or 6) Ability to withstand/avoid immune defences.

D. The increasing prevalence of antibiotic resistant bacteria has become a serious medical problem in recent years. Explain why the principle of selection is important in the emergence of antibiotic resistance. (1p)

Answer:

A mutation that makes the bacterium resistant to an antimicrobial drug will give the bacterium an advantage compared to other bacteria in the surroundings (eg in the gastrointestinal tract) only if its environment is exposed to that antibiotic. In that situation, susceptible bacterial in the environment will be inhibited, and the resistant strain will get access to nutrients previously utilized by other bacteria and can therefore multiply and dominate the microenvironment.

Question 4 (9p)

Cancer cells is the result of mutations in DNA that directly, and/or indirectly by epigenetic dysregulation, activates oncogenes and inactivates tumor suppressor genes. For a tumor to grow the crosstalk between the cancer cells and the tumor-microenvironment or stroma, which is constituted by various normal cell, also play an important role.

A. In 2011 Hanahan and Weinberg defined ten hallmarks of cancer, which all can be utilized as therapeutic targets. List at least five of these hallmarks. (2p)

5 of these:

Evading growth suppression	Activation of invasion and metastaisi
Avoiding immune detection	Inducing angiogenesis
Enabling replicative immortality	Genome instability
Tumor promoting inflammation	Resisting cell death/apoptosis
Deregulation of cellular metabolism	Sustained proliferative signalling (RTKs)

B. Multiple different cancer therapies exists. What is the principal difference between classical chemotherapy and targeted therapy? Give three examples of groups of targeted therapies. (2p) Chemotherapeutics target all dividing cells (cytotoxic, not very specific), targeted therapies aim to specifically inhibit specific (dysregulated) targets in the cancer cells (cytotoxic/cytostatic). E.g. Kinasei, PARPi, BETi, HDACi

C. Immunotherapy has become an integrated part of many cancer treatment regimes. What is the definition of immunotherapy? Give examples of immunotherapy. (2p) Therapy where the immune system is HELPED to recognize cancer. E.g. cytokines, checkpoint inhibition (anti CTLa-4, PD1 or PDL1-L2), BCG (bladder), antibodies, vaccination, Car-T

D. The EGFR-pathway is frequently mutated in multiple cancers, and this is important as it implicates multiple downstream pathways. Give examples of three cancer types with dysregulated EGFR signaling and how this pathway is inhibited in the different examples given. (3p)

Breast: anti-HER2 therapy, combination between anti-HER2 and EGFR-inhibition

Lung: Pan-EGFR inhibition, or downstream inhibitors such as BRAFi and MEKi

Colon: mABs to EGFR in KRAS wt tumors, combo of EGFR and BRAFi

MCQ

Only one letter (A,B,C, D or E) for each question (the best alternative). Make a table and hand in your answers together with the assay questions.

1.

PSI-Blast uses an iterative approach to searching, in order to increase sensitivity. This means that it uses previous results to improve sensitivity in the next iteration (i.e. next search).	
What can often be a source of error in such iterative approaches?	
A	The iteration stops because new results are too similar to previous results, and do not give novel information.
B	The iterations give too many hits, which will make the statistics fail.
C	Inclusion of an unrelated sequence from a previous iteration may lead to more accumulation of unrelated sequences in subsequent iterations.
D	We stop the iterations too early, because they become increasingly time consuming.
E	The iterations may introduce too many gaps in the alignments.

2.

Cardiovascular disease (CVD) is the number one cause of death globally.	
Which are the two types of CVDs that together are responsible for the majority of deaths.	
A	Cerebrovascular disease and Rheumatic heart disease
B	Coronary heart disease and cerebrovascular disease
C	Congenital heart disease and Peripheral arterial disease
D	Cerebrovascular disease and Congenital heart disease

3.

Gastric bypass is a common type of weight loss surgery. Interestingly, it often improves type 2 diabetes long before patients lose much weight. Many recent studies may help explain why.	
Which is the main molecular pathway?	
A	Dipeptidyl peptidase-4
B	Glucagon-like peptide-1 and/or peptide YY
C	Glucagon-like peptide-2 and/or gastric inhibitory peptide
D	Insulin and glucagon

4.

Insulin regulates many processes.	
Which of the following is not an effect of insulin?	
A	Reduced glycolysis
B	Increased glucose transport into muscle
C	Inhibition of lipolysis
D	Stimulation of glycogen synthesis
E	Inhibition of gluconeogenesis

5.

Defects in a single gene may sometimes lead to hereditary predisposition for cancer.	
Which of the following statements about this kind of cancer inheritance is the most correct?	
A	Hereditary cancer usually involves inheritance of an oncogene
B	Hereditary cancer usually involves inheritance of a defect tumour suppressor gene
C	Hereditary cancer is a typical example of recessive mode of inheritance
D	Hereditary cancer is a typical example of dominant mode of inheritance

6.

Many different types of mutations are seen in human disease	
Which of the following types are most commonly seen?	
A	Small deletions or insertions
B	Nucleotide substitutions
C	Larger deletions or insertions
D	Chromosome loss

7.

DNA replication is continuous on the leading strand and discontinuous on the lagging strand.	
In what direction does DNA replication occur on the leading- and lagging strand, respectively?	
A	5' to 3' and 3' to 5'
B	3' to 5' and 5' to 3'
C	5' to 3' and 5' to 3'
D	3' to 5' and 3' to 5'

8.

Epigenetic marks on DNA and histones are important to regulate transcription	
Which of the following enzymes plays an important role in preventing transcription?	
A	Histone acetyltransferase
B	TET (ten-eleven translocase) enzymes
C	Histone methyltransferase
D	Histone deacetylase

9.

Tumor microenvironment (TME) is important for cancer growth as well as the immune status of the tumor.	
Which factors are believed to be most important for regulation of the immune response towards the cancer?	
A	Levels of CD28-B7 family molecules, e.g PD-1 - PD-L1/L2
B	Levels of Lactate, HIF-1, IL-10
C	Levels of Cytotoxic T-cells versus MDSC (myeloid-derived suppressor cells)
D	Levels of PI3K signalling in tumor-associated T cells

10.

Mammalian cells have several different types of receptors.	
Which of the statements below are most correct?	
A	G-proteins are always organized in trimers
B	Activated receptor tyrosine kinases are always organized in homodimers
C	Checkpoint receptors are found only on T cells and antigen-presenting cells (APC)
D	Androgen receptors are activated and dimerize upon ligand binding

11.

Colon rectal cancer (CRC) is one of the leading cause of death in USA and Europe.	
Which of the statements below are most correct?	
A	CRC runs in family, i.e. it is for the most part inherited
B	The majority of CRCs are mutated in the APC gene
C	The majority of CRCs are mutated in EGFR
D	Lynch syndrome /HNPCC have reduced DNA repair activity

12.

Virus often cause infections in the respiratory tract.	
How do viruses replicate?	
A	Viruses replicate mostly in the same way as bacteria.
B	Viruses replicate by binary fission
C	Viruses depend on other cells for replication
D	Viruses replicate by infecting bacterial cells

13.

The most common modes of inheritance are autosomal dominant, autosomal recessive, and X-linked recessive modes	
What of the following statements is «most true» for an X-linked recessive condition?	
A	Affects males more frequently than females
B	Males are more severely affected
C	Is transferred from the mother
D	Is not transferred from father to daughters

14.

Fibroblasts can be reprogramed into pluripotent stem cells (iPSCs).	
OSKM factors used in reprograming of the somatic cells refer to:	
A	c-Myc, Klf4, SMAD, Oct4
B	Klf4, SMAD, miR-200c, Oct4
C	c-Myc, Klf4, Sox2, Oct4
D	LIN28, Klf4, VPA, Oct4